

CSS

Display properties

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>CSS</title>
    <style>
      h2 {
        display: inline-block;
        /* display: block; */
        /* display: inline; */
        border: 2px solid black;
        /* width: 200px;
           height: 200px; */
        padding: 20px;
      }

      #container {
        background-color: #333;
        width: 100%;
        height: 500px;
        display: flex;
        /* justify-content: flex-end; */
        /* justify-content: center; */
        /* justify-content: space-between; */
        /* justify-content: space-around; */
        /* justify-content: space-evenly; */
        /* align-items: flex-start; */
        /* align-items: flex-end; */
        /* align-items: center; */
        justify-content: center;
        align-items: center;
        gap: 20px;
      }

      #container div {
        width: 100px;
        height: 100px;
        background-color: aqua;
      }

      #grid-container {
```

```

        background-color: #333;
        height: 500px;
        margin-bottom: 20px;
        display: grid;
        /* grid-template-columns: 1fr 1fr 1fr 1fr; */
        grid-template-columns: repeat(4,1fr);
        grid-template-rows: 200px 200px;
        /* justify-items: end; */
        justify-items: center; /* horizontal */
        align-items:center;
        align-content:center;
        justify-content:center;
    }
    #grid-container div {
        width: 100px;
        height: 100px;
        background-color: aqua;
    }
</style>
</head>
<body>
    <h2>Cascading style sheet</h2>

    <div id="grid-container">
        <div></div>
        <div></div>
        <div></div>
        <div></div>
        <div></div>
        <div></div>
        <div></div>
        <div></div>
    </div>

    <div id="container">
        <div></div>
        <div></div>
        <div></div>
        <div></div>
    </div>
</body>
</html>

```

Interview-Based Theory

1. Display Property

- **block** → takes full width, starts on a new line.
- **inline** → fits content only, does not start on a new line.
- **inline-block** → behaves like inline but allows width/height adjustments.
- **none** → hides the element.

2. Flexbox

- **Definition:** One-dimensional layout system for arranging items in rows or columns.
- **Key properties:**
 - **display: flex** → activates flexbox.
 - **justify-content** → aligns items horizontally (flex-start, center, space-between, space-around, space-evenly).
 - **align-items** → aligns items vertically (flex-start, center, flex-end).
 - **gap** → spacing between items.

3. CSS Grid

- **Definition:** Two-dimensional layout system for rows and columns.
- **Key properties:**
 - **display: grid** → activates grid.
 - **grid-template-columns** → defines column structure (repeat(4, 1fr)).
 - **grid-template-rows** → defines row structure.
 - **justify-items** → horizontal alignment inside grid cells.
 - **align-items** → vertical alignment inside grid cells.
 - **align-content / justify-content** → alignment of the whole grid.

4. Flexbox vs Grid

Feature	Flexbox	Grid
Dimension	One-dimensional	Two-dimensional
Use case	Align items in a row/column	Complex layouts with rows + columns
Syntax	<code>display: flex</code>	<code>display: grid</code>

Coding Round Questions

- Display property → Explain difference between block, inline, and inline-block.
- Flexbox basics → Create a flex container with items aligned center.
- Grid basics → Create a grid with 4 columns and 2 rows.
- Flex vs Grid → Compare Flexbox and Grid with examples.
- Alignment → Show how to align items horizontally and vertically.
- Responsive design → Build a responsive layout using Flexbox and Grid.

⚡ Machine Coding Round Challenge

Problem: Dashboard Layout

Requirements:

- Create a header with navigation links.
- Use CSS Grid to create a 2x2 section for widgets.
- Use Flexbox to align items inside each widget.
- Add hover effects for interactivity.
- Ensure layout is responsive.

Previous class notes :

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Units</title>
    <style>
      /* h1{
        font-size: 60px; fixed size in all devices screen
      } */

      html {
        /* font-size: 20px; */
      }
      div {
        font-size: 15px;
        height: 50vh;
        background-color: aqua;
      }
    </style>
  </head>
  <body>
    <div>
      <h1>Units</h1>
    </div>
  </body>
</html>
```

```

    /* width: 50vw; */
    width: 50%;
    /* vw -> viewport width */
}

p {
    /* font-size: 2rem; */
    /* font-size: 2em; */
    /* text-align: center; */
    /* text-transform: lowercase; */
    /* text-transform: uppercase; */
    /* text-transform: capitalize; */
    /* color: yellow; */
    /* indentation */
    /* text-indent: 70px; */
    /* word-spacing: 20px; */
    /* letter-spacing: 10px; */
    /* line-height: 2; */
    /* text-shadow:10px 10px 5px blue ; */
    /* h-sh , v-sh, blur ,color */
    /* text-decoration-line: underline;
    text-decoration-color: red;
    text-decoration-thickness: 2px;
    text-decoration-style:double; */
    /* text-decoration: underline 20px red dotted; shorthand property */
    /* font-size:20px ; */
    border: 2px solid black ;
    padding: 10px;
    font-weight: 900;
    font-style: italic;
    font-family:
        "Franklin Gothic Medium", "Arial Narrow", Arial, sans-serif;
}

    /* vh = viewport height */
</style>
</head>
<body>
    <!-- units , text, font , box model -->

    <!-- text properties -->

    <!-- <del>Delete</del> -->

    <p>Lorem</p>
    <p>Lorem</p>
    <em>Lorem</em>

```

```

<h1>Units</h1>

<div>
  <p>Hello there 30px</p>
</div>

<span>span element</span>

<!-- units -> absolute, relative -->

<!-- absolute -> pixel -->
<!-- relative -> rem, em, vh, vw, % -->
</body>
</html>

```

```

<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>CSS</title>
    <style>
      /* universal selector */
      *{
        background-color: yellowgreen;
      }
      h1 {
        background-color: yellow;
      }

      /* element(type) selector */
      /* div{
        background-color: #333;
        width: 200px;
        height: 200px;
      } */

      /* #id selector */
      #idselector{
        font-size: 60px;
      }

      /* .class selector (group of elements) */

```

```

.item1{
    background-color: blueviolet;
}
/* group selector */
h3,h4,h5,.item1,#idselector{
    color: purple;
    background-color: antiquewhite;
}
/* attribute selector */

[title]{
    width: 200px;
    height: 200px;
}
[target="_blank"]{
    color: red;
}

/* descendent selector */
ol li{
    color: purple;
}

/* child selector */

nav ol li{

}
/*
div p{
    color: aqua;
} */
div>p{
    color: aqua;
}

/* + adjacent sibling selector */
/* h2 + p{
    color:red;
} */
/* general sibling selector ~ */

h2 ~ p{
    color: orangered;
}

```

```
</style>
<link rel="stylesheet" href="./demo.css">
</head>
<body>
  <p>Sibling below selector</p>
  <h2>Sibling selector</h2>
  <p>Sibling below selector</p>
  <p>Sibling below selector</p>

  <div>
    <p>child element</p>
    <section>
      <p>grand child element</p>
    </section>

    <p>child element</p>
  </div>

  <ol id="ol">
    <li>item 1 <li>item 1a</li> </li>
    <li>item 2</li>
    <li>item 3</li>
    <li>item 4</li>
    <li>item 5</li>
  </ol>
  <ul>
    <!-- <li class="item1">item 1</li> -->
    <li>item 1</li>
    <li>item 2</li>
    <!-- <li class="item1">item 3</li> -->
    <li>item 3</li>
    <li>item 4</li>
  </ul>

  <nav>
    <ol>
      <li>

      </li>
    </ol>
  </nav>

  <section>
    <ol>
      <li>
```



```
    </li>
  </ol>
</section>
```

```

```

```
<a href="#" target="_blank" >Blank</a>
<a href="#" target="_self" >Self</a>
```

```
<h3>H3</h3>
<h4>H4</h4>
<h5>H5</h5>
```

```
<!-- list -> ordered list, unordered list, description list -->
```

```
<h1 id="idselector">CSS</h1>
<h1 ">CSS</h1>
<div>
```

```
</div>
<div>
```

```
</div>
<!-- css -> 1994 -> hakon wium lie -->
```

```
<!-- html 5 -->
<!-- css -> css3 -->
```

```
<!-- include css -> inline(attribute), internal(tag), external -->
```

```
<h2 style="color: blue; background-color: #333">Hello there</h2>
```

```
<!--
```

```
    selector {
      property : value:
    } -->
  </body>
</html>
```

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Beginner Practice Template</title>
  </head>
  <body>
    <!-- Headings -->
    <h1>Main Heading (H1)</h1>
    <h2>Sub Heading (H2)</h2>
    <h3>Smaller Heading (H3)</h3>
    <h4>Heading H4</h4>
    <h5>Heading H5</h5>
    <h6>Heading H6</h6>

    <hr /> <!-- horizontal line to separate sections -->

    <!-- Paragraph -->
    <p>
      This is a sample paragraph. Students can edit this text to practice
      writing content inside the <code>&lt;p>&gt;</code> tag.
    </p>

    <br /> <!-- line break -->

    <!-- Links -->
    <a href="https://www.flipkart.com/" target="_blank">Visit Flipkart</a>
    <br />
    <a href="https://www.google.com" target="_blank">Visit Google</a>

    <hr />

    <!-- Images -->
    
    <br />
```

```

```

```
<hr />
```

```
<!-- Lists -->
```

```
<h2>Ordered List</h2>
```

```
<ol>
```

```
  <li>First item</li>
```

```
  <li>Second item</li>
```

```
  <li>Third item</li>
```

```
</ol>
```

```
<h2>Unordered List</h2>
```

```
<ul>
```

```
  <li>Apple</li>
```

```
  <li>Banana</li>
```

```
  <li>Cherry</li>
```

```
</ul>
```

```
<hr />
```

```
<!-- Table -->
```

```
<h2>Student Marks Table</h2>
```

```
<table border="1">
```

```
  <tr>
```

```
    <th>Name</th>
```

```
    <th>Math</th>
```

```
    <th>Science</th>
```

```
  </tr>
```

```
  <tr>
```

```
    <td>Alice</td>
```

```
    <td>90</td>
```

```
    <td>85</td>
```

```
  </tr>
```

```
  <tr>
```

```
    <td>Bob</td>
```

```
    <td>78</td>
```

```
    <td>88</td>
```

```
  </tr>
```

```
</table>
```

```
<hr />
```

```
<!-- Form -->
```

```
<h2>Simple Form</h2>
```

```
<form>
```

```
  <label for="name">Name:</label>
```

```
  <input type="text" id="name" /><br /><br />
```

```
<label for="email">Email:</label>
<input type="email" id="email" /><br /><br />

<label for="password">Password:</label>
<input type="password" id="password" /><br /><br />

<input type="submit" value="Submit" />
</form>
</body>
</html>
```